

Assignment-III on JAVA programming

1. WAP in Java to create and initialize a 1-D integer array with 10 random numbers between 1 and 100. Then, print the array elements.
2. WAP in Java to create and initialize a 3x3 2-D integer array. Calculate and print the sum of all elements, the sum of each row, and the sum of each column.
3. WAP in Java that takes an integer n as input from the user. Dynamically allocate a one-dimensional array of size n and fill it with Fibonacci numbers. Print the resulting array.
4. WAP in Java to concatenate the two arrays into a new array and return it. Print the elements of the concatenated array.
5. WAP in Java to perform the following operations:
 - Find the length of the string.
 - Convert the string to uppercase and lowercase.
 - Check if the string contains a specific substring (e.g., "Java").
 - Replace all occurrences of a character (e.g., 'a') with another character (e.g., 'b').
6. WAP in Java that demonstrates the immutability of strings. Create a string and perform some modifications on it. Print the string before and after the modifications. Also, demonstrate the difference between == and .equals() when comparing strings.
7. WAP in Java that takes a string as input from the user. Pass this string to a method that reverses the string. Print the reversed string in the main method.
8. WAP in Java that uses the StringBuffer class to perform the following operations on a string:
 - Append a string to the end of the buffer.
 - Insert a string at a specific position.
 - Delete a portion of the string.
 - Reverse the string.
9. WAP in Java that takes the user's name and age as input from the console using the **Scanner** class. Then, print a formatted output to the console, such as "Hello, [name]! You are [age] years old."